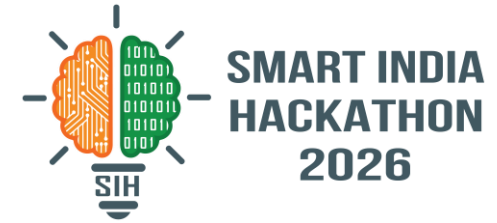


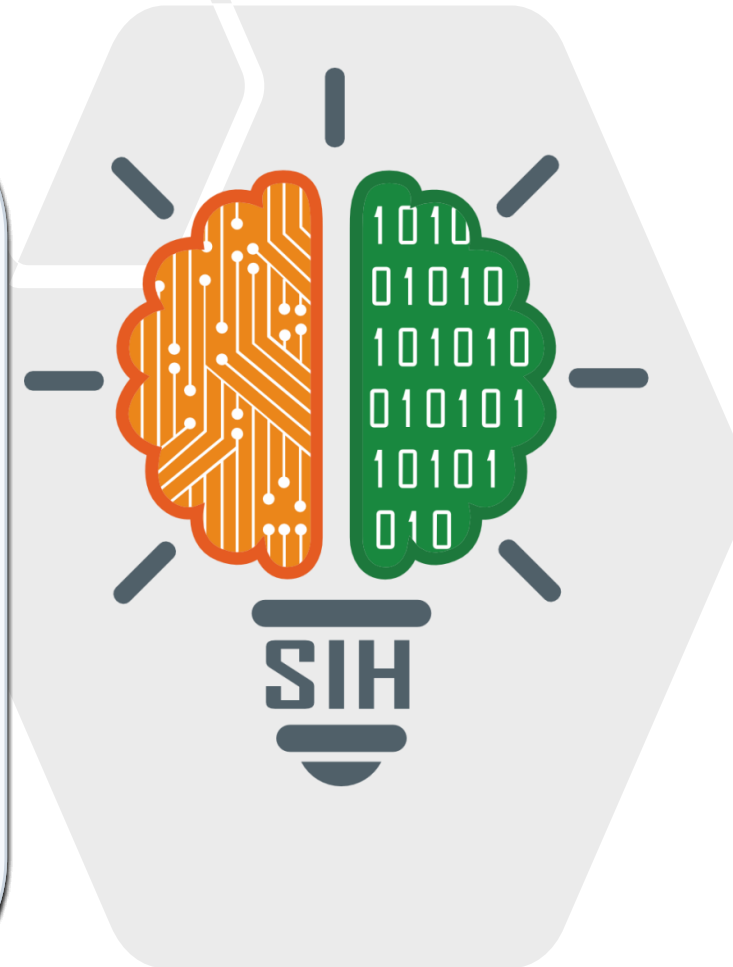
SMART INDIA HACKATHON 2026



TITLE PAGE: ZENITH

Autonomous Multi-Modal Lunar Image Registration & Sub-Pixel Photogrammetry

- **Problem Statement ID:** SIH26166
- **Problem Statement Title:** Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)
- **Theme:** Space Technology
- **PS Category:** Software
- **Team ID:** SIH26-A0H-T363
- **Team Name (Registered on portal):** Team Zenith





THE PROBLEM & WHY IS THIS HARD?

- **1. Sun-Angle / Illumination Shifts**

Extreme solar angle shifts cause complete crater shadow inversions (bright-to-dark gradient reversal across identical topography).

- **2. Extreme Scale Disparities**

Resolutions span 0.25 m (OHRC) → 5.0 m (TMC-2) → 80 m (IIRS) per pixel (up to 320x scale gap exceeding standard matchers).

- **3. Cross-Modal Sensor Physics**

Panchromatic optical vs. 256-band hyperspectral cubes with non-linear radiometric response functions.

- **Result of Conventional CV**

Standard SIFT/ORB algorithms fail, produce severe spatial clustering, or output high false-positive rates.

OUR SOLUTION: 6-STAGE ZENITH ARCHITECTURE

- ▶ **ADAPT:** Make source & reference imagery comparable using sensor, metadata, illumination & scale information.
- ▶ **REPRESENT:** Extract stable structural features via 2D Log-Gabor Phase Congruency & multi-scale pyramids.
- ▶ **MATCH:** Generate candidate correspondences via hybrid ensemble (RootSIFT + LoFTR deep transformer fallback).
- ▶ **VERIFY:** Use USAC_MAGSAC++ and geometric consistency to reject false matches and estimate 8-DOF Homography H.
- ▶ **REFINE:** Select spatially distributed reliable matches (4x4 grid) and refine locations to sub-pixel precision.
- ▶ **OUTPUT:** Return registered image (Lanczos-4), verified match points, RMSE, inlier ratio, coverage & confidence status.

WHY ZENITH? KEY INNOVATIONS

- ✓ **Physics & Metadata Aware:** Integrates PDS4 sensor parameters, sun angles, and PCA spectral reduction.
- ✓ **Illumination Robust:** Phase congruency extracts Fourier harmonic alignment invariant to lighting.
- ✓ **Hybrid Matcher Consensus:** Combines high-speed RootSIFT with deep LoFTR transformer fallback.
- ✓ **Spatially Uniform Matches:** 4x4 spatial grid balancing eliminates single-crater clustering.
- ✓ **Sub-Pixel Refinement:** Förstner gradient structure tensors achieve sub-pixel accuracy (< 0.5 px).
- ✓ **Confidence-Aware Output:** Decision Engine certifies result: REGISTERED / LOW_CONFIDENCE / FAILED.
- ✓ **Scientific Safety:** Strictly non-generative; raw registered output preserved with zero hallucination.



► **INPUT PAIR**

Source (OHRC / TMC-2 / IIRS) + Reference Base Map (LROC NAC / TMC DEM / Synthetic)

► **1. ADAPT**

Radiometric 1-99% quantile scaling, sensor bit-depth normalization, IIRS 256-band PCA

► **2. REPRESENT**

2D Log-Gabor Phase Congruency (Kovesi, 3 scales, 6 orientations) + Multi-Scale Pyramid

► **3. MATCH**

Multi-Matcher Ensemble: Primary RootSIFT (Hellinger kernel) + LoFTR Deep Transformer

► **4. VERIFY**

USAC_MAGSAC++ robust estimation of 8-DOF Homography H (adaptive noise $\sigma = 3.0$ px)

► **5. REFINE**

Spatial 4x4 Grid Inlier Balancing (prevents cluster collapse) + Förstner Structure Tensor

► **6. EVALUATE**

Decision Engine: Inlier Ratio ($\geq 35\%$), Spatial Coverage (≥ 0.35), Reprojection Residuals

► **7. OUTPUT**

Lanczos-4 Sub-Pixel Warping → Raw Registered Output + JSON Telemetry + Match Table

CORE DESIGN PRINCIPLE: AI / feature matching proposes correspondences → Geometry verifies them → Spatial balancing improves coverage → Refinement improves precision → Quality metrics decide whether the result is trustworthy.

IMPLEMENTATION TECHNOLOGY STACK

- **Core & Computer Vision:** Python 3.10+, OpenCV (USAC_MAGSAC++, Lanczos-4, CLAHE), NumPy, SciPy, scikit-image
- **Deep Learning & Matchers:** PyTorch, LoFTR Transformer (Local Feature TRansformer), Kornia, FLANN Matcher
- **Geospatial & Metadata:** GDAL, Rasterio, Planetary Data System (PDS4/LBL) parser, IAU Lunar Reference Models
- **Interface & Demonstration:** Streamlit Mission Control UI, Interactive Comparison Slider, Matplotlib, ReportLab

CHALLENGES & ZENITH TECHNICAL MITIGATION

- **Sun-Angle / Shadow Shifts:** 2D Log-Gabor Phase Congruency → Fourier harmonic phase alignment is completely contrast/brightness invariant.
- **Extreme Scale Gap (320x):** Multi-Scale Pyramids & Hierarchy → Multi-resolution octave decimation handles large scale differences progressively.
- **Cross-Modal Mismatch:** PCA Reduction + Structural Maps → Matches shared surface topography without requiring raw intensity equivalence.
- **Severe Outlier Matches:** USAC_MAGSAC++ Estimator → Marginalizes over continuous noise scale; tolerates up to 85% outlier rates.
- **Match Clustering Collapse:** 4x4 Spatial Grid Balancing → Enforces distributed inliers across 16 cells; prevents single-crater bias.
- **Low-Texture Maria Basins:** LoFTR Transformer Fallback → Global contextual cross-attention finds matches where local gradients fail.
- **Large Swath Computation:** Tile-Based Processing & FP16 → Memory-capped processing (< 1.4 GB VRAM) with fast CPU fallback.

CURRENT PROTOTYPE STATUS (DEMONSTRATED)

- ✓ **End-to-End Pipeline:** Verified across MVP1–MVP8 test runners with 100% test pass rate.
- ✓ **Measured Sub-Pixel RMSE:** 0.154 px (Nominal Optical), 0.730 px (Difficult Terminator Ramp).
- ✓ **Streamlit Web Interface:** Live Mission Control UI with interactive before/after split slider.
- ✓ **Feature Alignment:** Spatially distributed inlier correspondence vector visualization.
- ✓ **Raw Output Preserved:** Strictly non-generative; raw registered homography & artifacts exported.

FUTURE VALIDATION & DEPLOYMENT ROADMAP

- **Expanded Chandrayaan-2 Dataset:** Multi-orbit validation across wider lunar longitudes/latitudes.
- **DEM-Assisted Geometry:** Integration with LOLA/TMC DEMs for extreme 3D relief correction.
- **Space-Hardened FPGA Quantization:** INT8 execution targeted for onboard Terrain Relative Navigation.

PROJECTED DELIVERABLES & CAPABILITIES

- 1. Automated Lunar Registration:** Aligns difficult planetary images without relying entirely on manual ground control tie points.
- 2. Reliable Correspondences:** Automatically identifies geometrically consistent, outlier-free match points across disparate sensors.
- 3. Spatially Distributed Matches:** 4x4 grid balancing ensures correspondences span the full terrain footprint rather than clustering.
- 4. Sub-Pixel Refinement:** Gradient structure tensor optimization achieves sub-pixel localization accuracy (< 0.5 px).
- 5. Quality-Aware Output:** Delivers verified aligned image + full telemetry (RMSE, Inlier Ratio, Spatial Coverage, Confidence Status).
- 6. Reusable Planetary Framework:** Modular open architecture adaptable to future ISRO and international deep space exploration missions.

MULTI-SECTOR VALUE & BENEFITS

- ★ **Scientific Value:** Enables precise co-registration of morphology (OHRC), topography (TMC-2), and mineralogy (IIRS) for landing site characterization (Chandrayaan-3/4, Artemis, Lunar South Pole water-ice mapping).
- ★ **Operational Value:** Can significantly reduce repetitive manual image-alignment effort for orbital cartographers, accelerating baseline mosaic generation at planetary data centers.
- ★ **Technical Value:** Provides measurable, explainable, and certified registration quality with autonomous confidence gates (eliminates undetected silent alignment errors).
- ★ **Strategic & Educational:** Strengthens Indian space-technology capabilities in autonomous computer vision and remote sensing, demonstrating indigenous student innovation in planetary science.

PRIMARY RESEARCH & DATA SOURCES

[1] **ISRO / ISSDC** Chandrayaan-2 Data Explorer & Payload Documentation (OHRC 0.25m, TMC-2 5m, IIRS 80m), PRADAN Archive.

[2] **NASA PDS / LROC** Lunar Reconnaissance Orbiter Camera Data Archive & USGS Astrogeology ISIS Documentation.

[3] **Kovesi, P. (1999/2003)** "Image Features From Phase Congruency," Videre / DICTA (Illumination-invariant harmonic model).

[4] **Lowe, D. G. (2004)** "Distinctive Image Features from Scale-Invariant Keypoints (SIFT)," IJCV.

[5] **Arandjelović & Zisserman (2012)** "Three things everyone should know to improve object retrieval (RootSIFT)," CVPR.

[6] **Sun, J. et al. (2021)** "LoFTR: Detector-Free Local Feature Matching with Transformers," IEEE/CVF CVPR.

[7] **Barath, D. et al. (2020)** "MAGSAC++, a Fast, Reliable and Accurate Robust Estimator," IEEE/CVF CVPR.

[8] **Li, J. et al. (2020)** "RIFT: Multi-Modal Image Matching Based on Radiation-Variation Insensitive Feature Transform," IEEE TIP.

CAPABILITY & BENCHMARK COMPARISON MATRIX

Capability	SIFT	LoFTR	RIFT	ZENITH
Illumination / Shadow Invariance	X Fails	△ Partial	✓ High	✓ Superior (Phase Congruency)
Scale Disparities (> 2x)	△ Moderate	X Weak	△ Moderate	✓ High (Pyramid+Ensemble)
Cross-Modal / Hyperspectral	X None	△ Limited	✓ Good	✓ Robust (PCA+Structural)
Sub-Pixel Refinement	X Coarse	△ Approx.	X None	✓ Structure Tensor (<0.5px)
Robust Outlier Rejection	X RANSAC	X RANSAC	X RANSAC	✓ USAC_MAGSAC++
Spatial Match Distribution	X Clustered	X Clustered	X Clustered	✓ 4x4 Grid Balancing
Autonomous Quality Gate	X None	X None	X None	✓ Decision Engine (3-State)

Legend: ✓ = Supported / High · △ = Partial · X = Unsupported / Fails